

Calculus II

Lecture 07

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Six important integrals

- ▣ **These integrals are follows**

$$1. \int \frac{dx}{\sqrt{a^2 - x^2}}$$

$$2. \int \sqrt{a^2 - x^2} \, dx$$

$$3. \int \frac{dx}{\sqrt{a^2 + x^2}}$$

$$4. \int \sqrt{a^2 + x^2} \, dx$$

$$5. \int \frac{dx}{\sqrt{x^2 - a^2}}$$

$$6. \int \sqrt{x^2 - a^2} \, dx$$

We first evaluate

$$\square \quad 1. \int \frac{dx}{\sqrt{a^2 - x^2}}$$

Put $x = a \sin \theta \Rightarrow dx = a \cos \theta d\theta$

$$\therefore \int \frac{dx}{\sqrt{a^2 - x^2}} = \int \frac{a \cos \theta d\theta}{\sqrt{a^2 - a^2 \sin^2 \theta}}$$

$$\int \frac{a \cos \theta d\theta}{a \cos \theta} = \int d\theta = \theta = \sin^{-1} \frac{x}{a}$$

$$2. \int \sqrt{a^2 - x^2} \, dx$$

Put $x = a \sin \theta \Rightarrow dx = a \cos \theta \, d\theta$

$$\int \sqrt{a^2 - x^2} \, dx = \int \sqrt{a^2 - a^2 \sin^2 \theta} \, a \cos \theta \, d\theta$$

$$= \int a \sqrt{1 - \sin^2 \theta} \cdot a \cos \theta \, d\theta = \int a^2 \cos^2 \theta \, d\theta$$

$$= a^2 \int \cos^2 \theta \, d\theta = a^2 \left[\frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right] + c$$

$$= a^2 \left[\frac{1}{2} \theta + \frac{1}{4} (2 \sin \theta \cos \theta) \right] + c = \frac{a^2}{2} \theta + \frac{a^2}{2} \sin \theta \cos \theta + c$$

$$= \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + \frac{a^2}{2} \left(\frac{x}{a^2} \right) \sqrt{a^2 - x^2} + c$$

$$= \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + \frac{x}{a} \sqrt{a^2 - x^2} + c$$

Now we evaluate

$$3. \int \frac{dx}{\sqrt{a^2 + x^2}}$$

Put $x = a \sinh \theta \Rightarrow dx = a \cosh \theta d\theta$

$$\begin{aligned} \therefore \int \frac{dx}{\sqrt{a^2 + x^2}} &= \int \frac{a \cosh \theta d\theta}{\sqrt{a^2 + a^2 \sinh^2 \theta}} \\ &= \int \frac{a \cosh \theta d\theta}{a \cosh \theta} = \int d\theta = \theta = \sinh^{-1} \frac{x}{a} \end{aligned}$$

Hence $\int \frac{dx}{\sqrt{a^2 + x^2}} = \sinh^{-1} \frac{x}{a}$

$$4. \int \sqrt{a^2 + x^2} dx$$

Put $x = a \sinh \theta \Rightarrow dx = a \cosh \theta d\theta$ [**$\cosh^2 \theta - \sinh^2 \theta = 1$**]

$$= \int \sqrt{a^2 + x^2} dx = \int \sqrt{a^2 + a^2 \sinh^2 \theta} a \cosh \theta d\theta$$

$$= \int a^2 \cosh^2 \theta d\theta = \frac{a^2}{2} \int (\cosh 2\theta + 1) d\theta = \frac{a^2}{2} \left[\frac{\sinh 2\theta}{2} + \theta \right]$$

$$= \frac{a^2}{2} [\sinh \theta \cosh \theta + \theta] = \frac{a^2}{2} [\sinh \theta (\sqrt{1 + \sinh^2 \theta}) + \theta]$$

$$= \frac{a^2}{2} \left[\frac{x}{a} \sqrt{1 + \frac{x^2}{a^2}} + \sinh^{-1} \frac{x}{a} \right]$$

Hence $\int \sqrt{a^2 + x^2} dx = \frac{a^2}{2} \left[\frac{x}{a} \sqrt{1 + \frac{x^2}{a^2}} + \sinh^{-1} \frac{x}{a} \right]$

Finally, we evaluate

5. $\int \frac{dx}{\sqrt{x^2 - a^2}}$

Put $x = a \cosh \theta \Rightarrow dx = a \sinh \theta d\theta$

$$\therefore \int \frac{dx}{\sqrt{x^2 - a^2}} = \int \frac{a \sinh \theta d\theta}{\sqrt{a^2 \cosh^2 \theta - a^2}} = \int \frac{a \sinh \theta d\theta}{a \sinh \theta} = \int d\theta = \theta$$

$$= \cosh^{-1} \frac{x}{a}$$

$$\text{Hence } \int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1} \frac{x}{a}$$

Class Work

6. $\int \sqrt{x^2 - a^2} \, dx$

Thank You

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